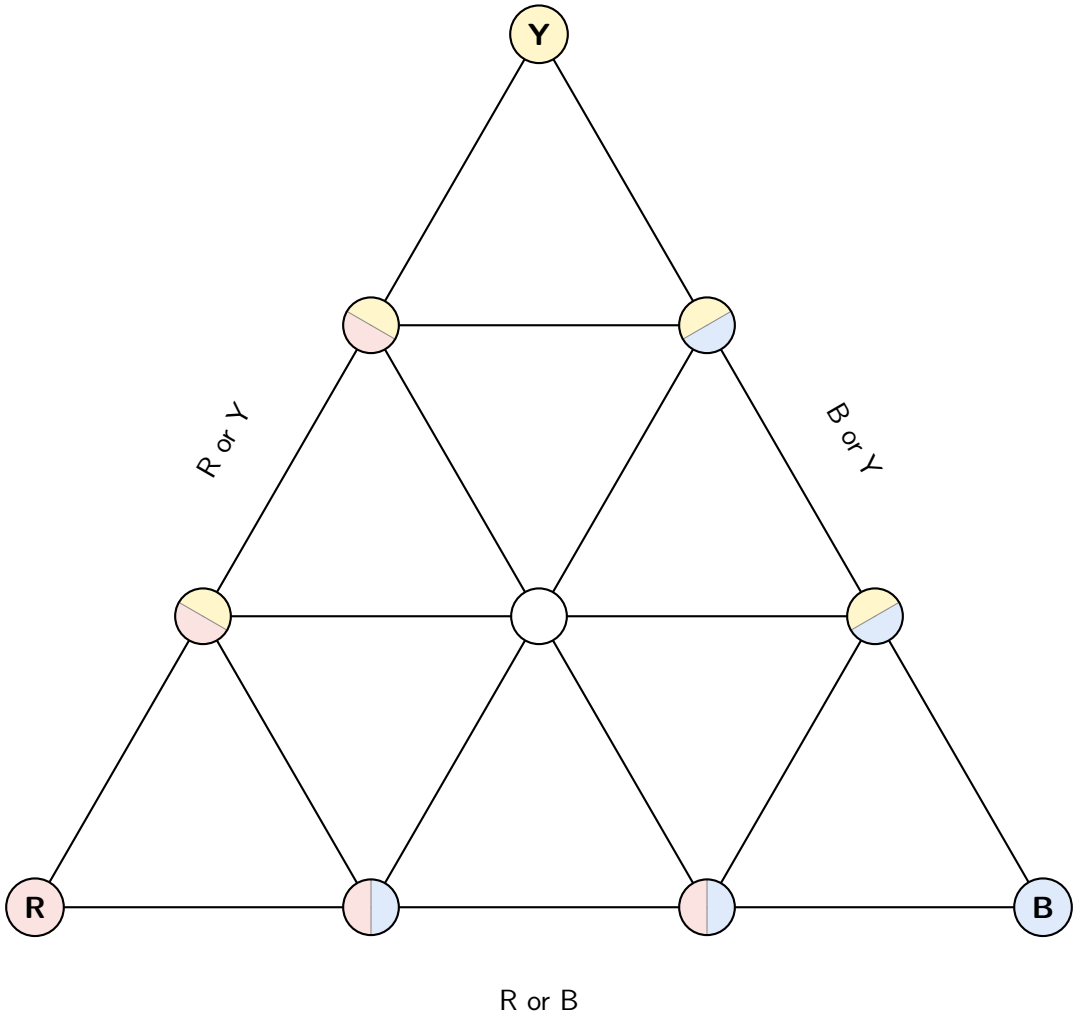
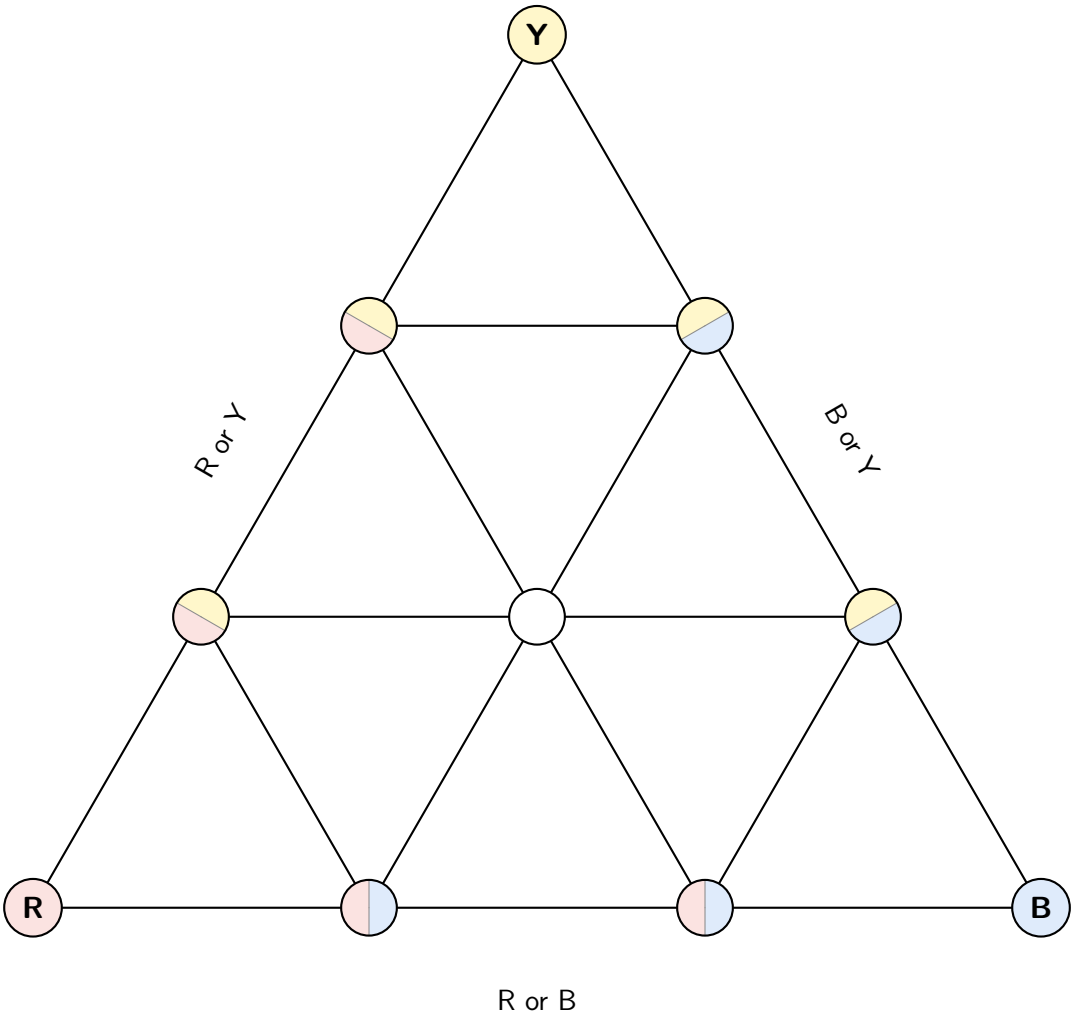
Put R, B, or Y on every empty dot. Keep the printed letters. Each outside side allows only the letters beside it. A little triangle has no lines inside.

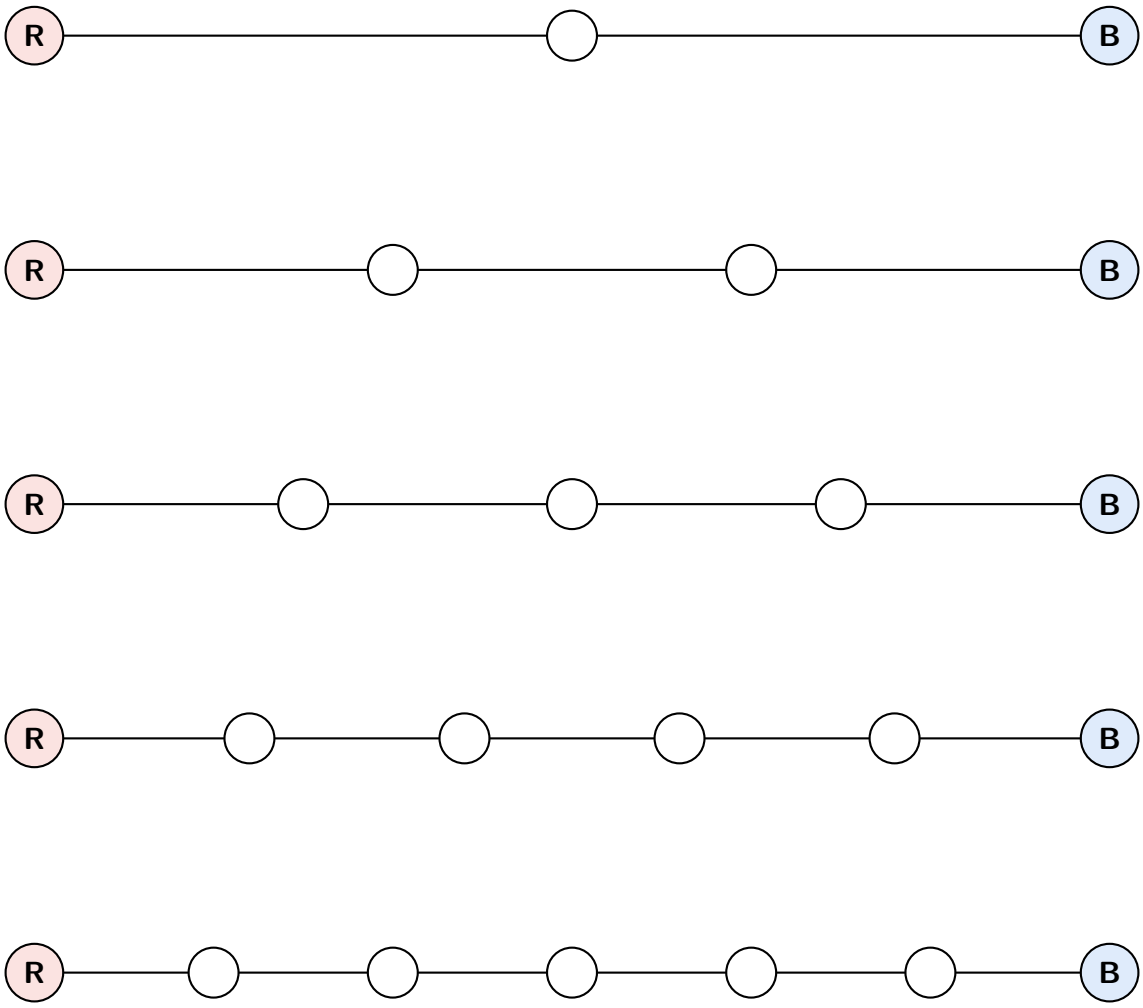
Problem 1: Try fillings with R, B, and Y counters. Find a filling with as few little triangles containing R, B, and Y as you can, and another with as many as you can. Write each record on its own board.



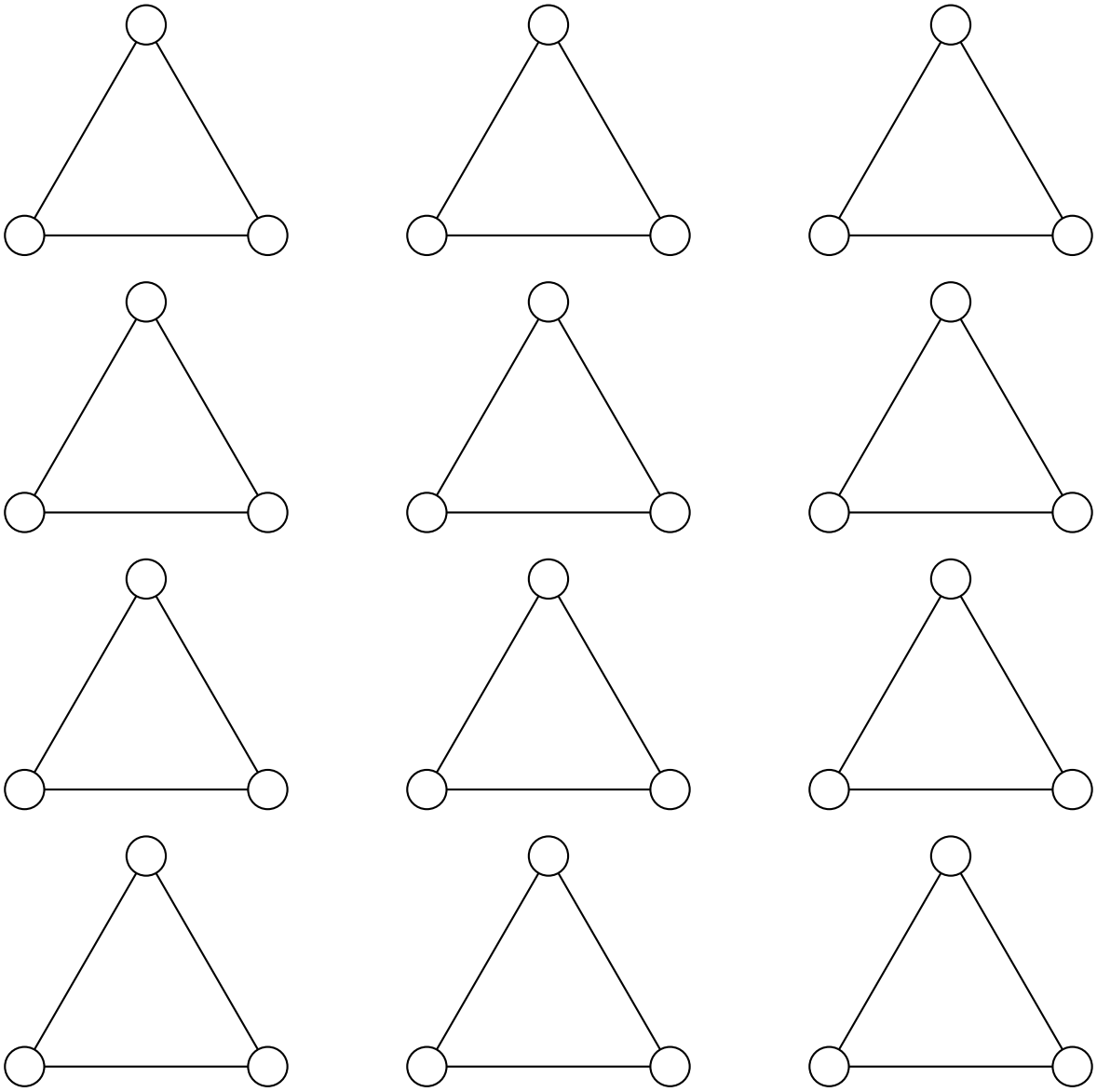
Another board for Problem 1.



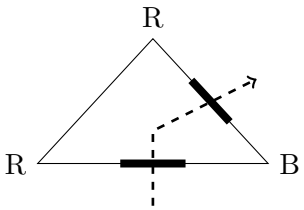
Problem 2: An edge with R at one end and B at the other is a door. Write R or B on every empty dot and mark the doors. Can any row have an even number of doors? Explain why or why not.



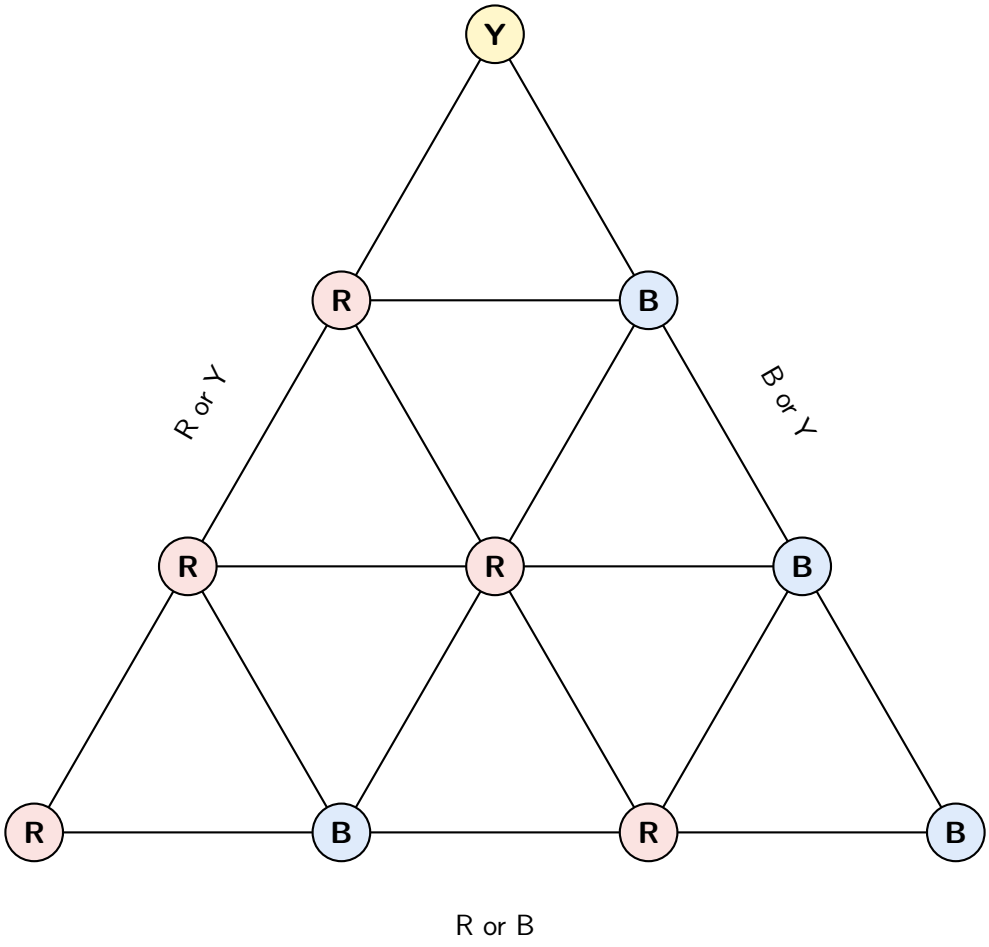
Problem 3: Fill these triangles with R, B, and Y in different ways, and mark the doors in each. Which fillings have exactly one door? Could a triangle have three doors?



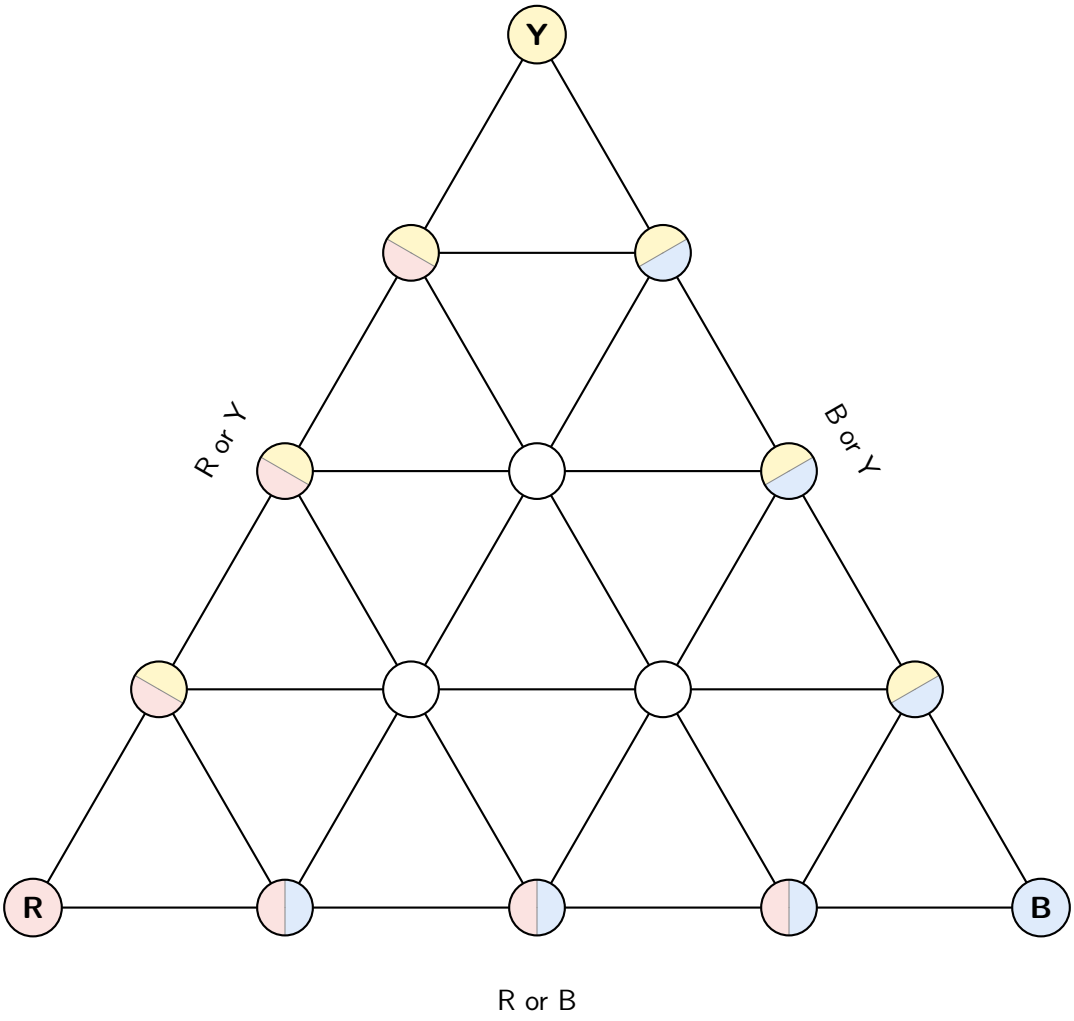
Problem 4: Mark every door. Draw every route that starts at an outside door. A route must leave a two-door triangle through its other door; it stops at a one-door triangle or outside. Which outside doors lead back outside?



Example: enter one thick door and leave through the other.
The dashed route crosses edges; it does not follow them.



Problem 5: Fill the dots so that exactly three little triangles have R, B, and Y. Can all the routes that start at doors on the bottom side end outside?



Problem 6: Can the dots be filled so that no little triangle has R, B, and Y? Give a reason that covers every filling allowed by the side rules.

